MAT1301

Made with Goodnotes

$(i_0 \cdots i_{n-1}) \in S_{(0,\dots,n-1)}$

$\wr$

$(i, j_0 \cdots j_{n-1}) \in S_{(0\dots n)}$ where $j_k = i_k$ if $i_k < i$, $j_k = i_k + 1$ if $i_k \geq i$

e.g. $n=3$, $i=1$: $\begin{array}{ccc} i & i_1 & i_2 \\ (1 & 2 & 0) \end{array}$ $\rightsquigarrow$ $t_0 = s_0$, $t_1 = 0$, $t_2 = s_1$, $t_3 = s_2$

$0 \leq s_1 \leq s_2 \leq s_0$

$t_1 \leq t_2 \leq t_3 \leq t_0$

$\begin{array}{cccc} (1 & 2 & 3 & 0) \\ i & j_0 & j_1 & j_2 \end{array}$

$i_0 \quad i_1 \quad i_2$

$(2 \quad 1 \quad 0)$

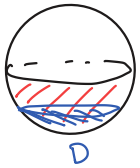
$0 \leq s_2 < s_1 \leq s_0$

$\begin{array}{cccc} \| & \| & \| & \| \\ t_1 & t_3 & t_2 & t_0 \end{array}$

$\begin{array}{cccc} (1 & 3 & 2 & 0) \\ j_0 & j_1 & j_2 & \\ \| & \| & \| & \\ i_0+1 & i_1+1 & i_2 & \end{array}$

as $i_0, i_1 \geq i$.

$H_i(S^n) \cong \begin{cases} \mathbb{Z} & i = 0, n \\ 0 & \text{else} \end{cases}$ we know $H_0(S^n) = \mathbb{Z}$.



$0 \to C_i(D^n_-) \to C_i(S^n) \to C_i(S^n)/C_n(D^n_-) \to 0$ $\qquad H_i(D^n) = \begin{cases} \mathbb{Z} & i=0 \\ 0 & \text{else} \end{cases}$

$\wr$

$0 \to H_i(S^n) \xrightarrow{\sim} H_i(S^n, D^n_-) \to 0$ $\quad i>1$, $\quad 0 \to H_i(S^n) \hookrightarrow H_i(S^n, D^n_-)$ $i=1$.

excision: $H_i(S^n, D^n_-) \xleftarrow{\sim} H_i(S^n \setminus D, D^n_- \setminus D)$

$\| \wr$

$H_i(D^n_-, S^{n-1}) \rightsquigarrow 0 \to C_i(S^{n-1}) \to C_i(D^n) \to C_i(D^n)/C_i(S^{n-1}) \to 0$

$\wr$

$0 \to H_i(D^n, S^{n-1}) \to H_{i-1}(S^{n-1}) \to 0$ $\quad i>1$. $\quad i=0$: know $H_0(D^n, S^{n-1}) = 0$.

$i=1$: $H_1(D^n) \to H_1(D^n, S^{n-1})$, $H_1(D^n) = 0$, $\to H_0(S^{n-1}) \xrightarrow{\sim} H_0(D^n) \to 0$, $H_0(S^{n-1}) = \mathbb{Z}$, $H_0(D^n) = \mathbb{Z}$

So $H_1(D^n, S^{n-1}) = 0$.

So $i>1 \Rightarrow H_i(S^n) \cong H_i(S^n, D^n) \cong H_i(D^n, S^{n-1}) \cong H_{i-1}(S^{n-1})$. $i>1$; 0 $i=0,1$

reduced is easier: pair (S^n, D^n_-) gives

$\cdots \to \underset{\substack{\| \\ 0 \text{ for all } i}}{\tilde{H}_i(D^n)} \to \tilde{H}_i(S^n) \to \tilde{H}_i(S^n, D^n_-) \to \underset{\substack{\| \\ 0 \text{ for } i>0}}{\tilde{H}_{i-1}(D^n)} \to \cdots$

So $\tilde{H}_i(S^n) \cong \tilde{H}_i(S^n, D^n)$ $\quad i>0$

excision: $\tilde{H}_i(S^n, D^n) \cong \tilde{H}_i(S^n \setminus D, D^n \setminus D) \cong \tilde{H}_i(D^n, S^{n-1})$

pair (D^n, S^{n-1}) gives

$\cdots \to \underset{\substack{\| \\ 0 \text{ all } i}}{\tilde{H}_i(D^n)} \to \tilde{H}_i(D^n, S^{n-1}) \to \tilde{H}_{i-1}(S^{n-1}) \to \underset{\substack{\| \\ 0 \quad i>0}}{\tilde{H}_{i-1}(D^n)} \to \cdots$

So $\tilde{H}_i(S^n) \cong \tilde{H}(S^n, D^n_-) \cong \tilde{H}_i(D^n, S^{n-1}) \cong \tilde{H}_{i-1}(S^{n-1})$ $\quad i>0$. we know $\tilde{H}_i(S^0) = \begin{cases} \mathbb{Z} & i=0 \\ 0 & i>0 \end{cases}$

So $\tilde{H}_i(S^1) = \begin{cases} \mathbb{Z} & i=1 \\ 0 & \text{else} \end{cases}$

$\vdots$

$\tilde{H}_i(S^n) = \begin{cases} \mathbb{Z} & i=n \\ 0 & \text{else} \end{cases}$

$\Downarrow$

$H_i(S^n) = \begin{cases} \mathbb{Z} & i=n,0 \\ 0 & \text{else} \end{cases}$.

$[v_0 \cdots v_n]$ n-simplex $\rightsquigarrow$ barycentric subdiv into n-simplices

$[b, w_0 \cdots w_{n-1}]$ where b is barycenter $\sum_{i=0}^{n} \frac{1}{n+1} v_i$ and

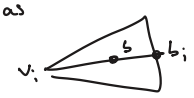
$[w_0 \cdots w_{n-1}]$ is $(n-1)$-simplex in barycentric subdiv. of $[v_0 \cdots \hat{v}_i \cdots v_n]$.

$\operatorname{diam}[b, w_0 \cdots w_{n-1}] \leq \frac{n}{n+1} \operatorname{diam}[v_0 \cdots v_n]$.

by induction $d(w_i, w_j) \leq$ ✓

remains: $d(b, v_i) \leq \frac{n}{n+1} \operatorname{diam}[v_0 \cdots v_n]$

as



$b = \frac{1}{n+1} v_i + \frac{n}{n+1} b_i$

So $d(b, v_i) = \frac{n}{n+1} d(b_i, v_i) \leq \frac{n}{n+1} \operatorname{diam}[v_0 \cdots v_n]$.

$S[\emptyset] = [\emptyset]$

$S[v_0 \cdots v_n] = b\, S\partial[v_0 \cdots v_n]$

chain map by induction: $\partial S[v_0 \cdots v_n] = \overset{1 - b\partial}{\partial b} S\partial[v_0 \cdots v_n]$

$= S\partial[v \cdots] - b\partial S\partial[v \cdots]$

$= S\partial[v \cdots]$ by ind.

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chain htpy $T_{-1} = 0$

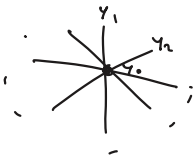
$$T\lambda = b(\lambda - T\partial\lambda)$$

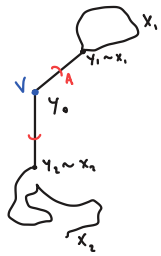
$$\begin{aligned}
\partial T\lambda &= \partial b(\lambda - T\partial\lambda) \\
&= (1 - b\partial)(\lambda - T\partial\lambda) \\
&= \lambda - T\partial\lambda - b(\partial\lambda - \partial T\partial\lambda) \\
&= \lambda - T\partial\lambda - b(S\partial\lambda - T\partial^2\lambda) \\
&= \lambda - T\partial\lambda - S\lambda
\end{aligned}$$

$\Rightarrow \partial T + T\partial = 1 - S.$

General chains

7.2. $(X_i, x_i)_{i=1}^n$ cnctd based spaces $\rightsquigarrow BM((X_i, x_i)) := Y \cup \bigsqcup_{i=1}^n X_i \big/ x_i \sim y_i$ where $Y =$





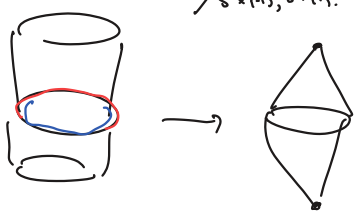
$\bar{V} \subset A^\circ$

$H_n(X \setminus V, A \setminus V) \xrightarrow{\sim} H_n(X, A)$

7.4. - $H_n(T^2)$

7.5. $f: S^n \longrightarrow S^n \rightsquigarrow \Sigma f: S^{n+1} \longrightarrow S^{n+1}$

$\Sigma S^n = S^n \times [-1,1] \big/ S^n \times \{-1\}, S^n \times \{1\}.$



7.6. Hom. $f: S^n \longrightarrow S^n \rightsquigarrow f(x) = \pm x$ for some x.

Lem. $f, g: S^n \longrightarrow S^n$ s.t. $f(x) \neq -g(x)$ $\forall x \in X$ then $f \simeq g$.

$h: S^n \times I \longrightarrow S^n$

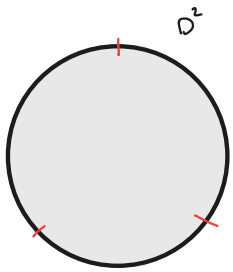
$(x, t) \longmapsto \dfrac{t f(x) + (1-t) g(x)}{\| \qquad \|} \neq 0$ as $t f(x) + (1-t) g(x) = 0$

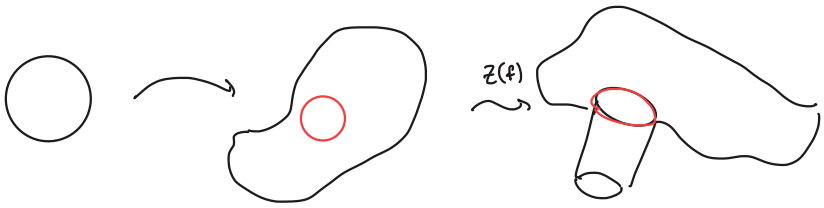
$\Rightarrow t f(x) = (t-1) g(x)$

$f: X \longrightarrow Y$

$Z(f) := X \times I \cup Y \big/ f(x) \sim (x, 0)$

$$\begin{tikzcd}
x \arrow[d, mapsto] & X \arrow[r, "f"] \arrow[d] & Y \arrow[d] \\
(x,0) & X \times I \arrow[r] & Z(f).
\end{tikzcd}$$





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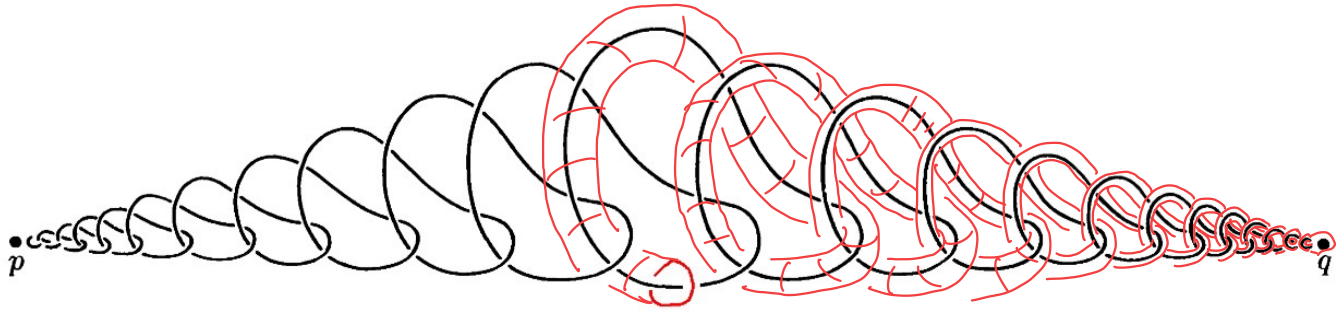
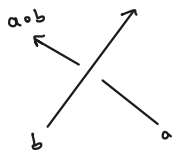


Knot quandle

Γ finite set. $\circ : \Gamma \times \Gamma \to \Gamma$ st.

1) $a \circ a = a$ (R1)
2) $\forall a, b \in \Gamma \ \exists! x : x \circ a = b$ (R2)
3) $(a \circ b) \circ c = (a \circ c) \circ (b \circ c)$ (R3)

a proper colouring of D (link diagram of oriented link K) is an assignment of a colour (element of Γ) to each arc st.



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